CMS RFI

https://www.federalregister.gov/documents/2025/05/16/2025-08701/request-for-information-health-technology-ecosystem

B. Patients and Caregivers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve patients and caregivers. While we certainly want patients and caregivers to answer questions in this section (and in other sections) from the patient/caregiver point of view, we also invite all stakeholders to provide their viewpoints on the patient/caregiver workflows.

1. Patient Needs

PC-1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?

As someone who could be considered more "tech savvy" than some of my loved ones, I would appreciate an app that would make scheduling appts, sending reminders, tracking healthcare notes, medications, and all other needs accessible to me online. Something that would allow me to access my own "EMR" of sorts. This would allow me and my loved ones to be more up to date and have better health literacy. For example, the U has an app called MyChart where you can see every med you take, all past appt notes, upcoming appts, scheduling, billing, and messaging. I've seen this app help patients be more health literate and advocate for themselves because they know what is going on.

a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?

Medication management, appointment scheduling and tracking, health history, recent labs, communication between providers, and links to health information that is applicable to the patient.

b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with? In your response, please consider tasks that could be supported or facilitated by software solutions in the future.

Billing would be top priority for me because it's difficult to navigate and understand. I would also like help with medication management with indications, when, how, what I take, etc.. I like to rely on them to schedule appointments when I need them because it's hard to keep track.

PC-2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?

Not my relatives, but I have a patient portal for myself.

a. If so, what are some examples of benefits it has provided?

I appreciate being able to see billing information and providers notes during my appts. I know many patients use it to track medication.

b. If not, in what contexts or for what workflows would it be most valuable to use one portal or system to access all such health information?

Keeping everything in one easy spot is nice for accessibility.

c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?

Sometimes apps or websites can feel like navigating a maze. Having access or knowing where to find labs, x-rays, notes, etc can be a challenge if it is not in an obvious, easy to see, way.

PC-3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?

Apps like MyChart from the University of Utah

PC-4. What features are missing from apps you use or that you are aware of today?

Direct communication with providers or access to treatment plans.

a. What apps should exist but do not yet? Why do you believe they do not exist yet?

Apps that could keep track of patient's medical records, but it’s hard to keep a universal system.

b. What set of workflows do you believe CMS is uniquely positioned to offer?

Possibly an app like MyChart where patients can have access to all healthcare related needs and information. On top of that, providing accessible health information that can help navigate diagnoses or treatments.

PC-5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?

Improved health literacy, compliance, medical safety, and overall safer and better health outcomes for all involved parties, which would also be more cost effective in the long run.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?

CMS should provide data security and patient privacy within the app in compliance with HIPAA laws and government entities, ensuring that only authorized individuals can access sensitive health information. This app should also address health disparities and improve access to care for underserved populations and be accessible to individuals with diverse needs and abilities. It should be clear and easy to navigate.

b. What changes would enable timely access to high quality CMS and provider generated data on patients?

Universal health system for patient portals

PC-6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?

Clear tabs/menu options, notifications, calendar tool, not needing to click through multiple tabs to find something.

PC-7. If CMS were to collect real-world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?

To get patient's thoughts, a survey to assess opinions on using a digital health product. For example, “On a scale of 1-10, 10 being most knowledgeable, how do you feel like using this product improved your healthcare knowledge” or “This product made healthcare more accessible for me, agree, disagree”.

2. Data Access and Integration

PC-8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?

Paperwork can be helpful to give information about medication, diagnosis, and billing, on both sides.

a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?

Billing information (itemized billing information), encounter notes, medications, prices, claims

b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?

Medication management so patients know what they are taking.

c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?

A universal data tool where every entity that provides healthcare can access and chart so patients can see information from every visit.

PC-9. Given that the Blue Button 2.0 API only includes basic patient demographic, Medicare coverage, and claims data (Part A, B, D), what additional CMS data sources do developers view as most valuable for inclusion in the API to enable more useful digital products for patients and caretakers?

Medication management, chart reviews, eligibilities

a. What difficulties are there in accessing or utilizing these data sources today?

Different healthcare facilities chart differently, and there is no uniform online patient portal to use.

b. What suggestions do you have to improve the Blue Button 2.0 API experience?

Add more clinical information

c. Is there non-CMS data that should be included in the API?

Medication management and appointment summaries

PC-10. How is the Trusted Exchange Framework and Common AgreementTM (TEFCATM) currently helping to advance patient access to health information in the real world?

It serves as a national framework that connects these separate healthcare networks, creating a single, secure way for health information to be shared across the country.

a. Please provide specific examples.

Both the California hospital and the Utah doctor are connected to different "Qualified Health Information Networks" (QHINs), which are now linked by TEFCA. The Utah doctor can instantly and securely request the patient's full electronic record from the California hospital through their QHIN.

b. What changes would you suggest?

Provide stronger financial incentives, grants, or technical assistance for smaller practices, rural hospitals, and independent providers to join QHINs.

c. What use cases could have a significant impact if implemented through TEFCA?

When a patient is unconscious or unable to communicate, TEFCA would allow clinicians to instantly access their complete medical history, including allergies, current medications, pre-existing conditions, past surgeries, and emergency contacts, regardless of where the patient has previously received care.

d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?

None that I am aware of.

e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?

None that I am aware of.

f. Are there redundant standards, protocols, or channels that should be consolidated?

Many healthcare groups use two different ways to share data (HL7 v2 and FHIR) or find it hard to switch from the old way. If everyone just used FHIR as the main standard, it would make building new systems much easier, cut down on costs, and unlock data for more powerful apps and analysis.

g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?

Nothing widespread.

PC-11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?

These systems help healthcare teams to get a full picture of a patients health quickly, leading to better coordinated, accessible, and safer care.

a. How valuable, available, and accurate do you find the data they share to be?

Sometimes they transfer information well but it can be hard to access. It may take longer to find information rather than just calling the facility.

b. What changes would you suggest?

Ensure the system clearly indicates data location and mandate the timely upload of accurate information.

c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?

Care Everywhere can be a helpful tool that allows Epic EMRs to see other facility notes.

d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?

HIEs continue to be vital as regional connectors, linking local doctors, hospitals, and smaller clinics that national networks might miss. They act as neutral, trusted hubs for sharing data within communities, helping with public health and offering specialized local services.

PC-12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?

Making real-time, personalized cost estimates and easily viewing provider availability for quick booking would greatly simplify care navigation. Provide clear patient-facing quality data and ensuring seamless data transfer during referrals.

a. Examples may include the following:
    1) Binding cost estimates for pre-defined periods.
    2) Viewing provider schedule availability.
    3) Using third-party apps for appointment management.

4) Accessing patient-facing quality metrics.
5) Finding the right provider for specific healthcare needs.

a. What use cases are possible today?
b. What should be possible in the near future?
c. What would be very valuable but may be very hard to achieve?

3. Information Blocking and Digital Identity

PC-13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?

Launch clear public awareness campaigns and making the reporting process super simple. This would directly empower individuals to ensure their data access rights are honored. Increasing these complaints would ultimately advance data exchange by holding those who unfairly block information accountable, forcing better sharing practices across the healthcare system.

PC-14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 credentialing service providers (CSP)):

a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?

Getting patients and caregivers to use strong digital identity credentials is hard because the sign-up process is often complicated and confusing. Many people don't see enough immediate benefit to justify the effort, and they have privacy worries about sharing sensitive health data. Plus, these credentials aren't yet accepted everywhere, so patients still need multiple logins.

b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?

If widely used, digital identity credentials would mean patients and caregivers could securely and easily access all their health information and services with just one trusted login. This would drastically reduce the hassle of multiple passwords and paper forms, making care navigation smoother and more efficient. Ultimately, it enhances both security and patient control over sensitive health data.

c. What are the potential downsides?

It could unfairly exclude those without tech access, like the elderly or low-income. It also creates a huge target for data breaches, risking a person's entire health profile. There are also concerns about losing control over personal data and the complexity of managing access for caregivers, along with the risk of system failures that could block access to care.

d. How would encouraging the use of CSPs improve access to health information?

This would simplify how patients access their health information. Instead of managing many different usernames and passwords for various doctors and hospitals, people could use one secure, verified login. This consistent, trusted method would make getting and sharing health data much easier and safer for everyone.

e. What role should CMS/payers, providers, and app developers have in driving adoption?

CMS and other payers should incentivize and eventually require the use of these digital IDs, ensuring consistent standards for their acceptance. Healthcare providers must integrate these logins into their systems and actively educate patients on their benefits. App developers need to create user-friendly and valuable apps that truly leverage these secure identities to make patient health management easier.

f. How can CMS encourage patients to get digital identity credentials?

CMS can encourage patients to get digital identity credentials by clearly showing them how it makes managing their Medicare benefits easier and safer, perhaps through targeted campaigns. They should also integrate it seamlessly into existing services, making it the most convenient way to access their health information and reducing any sign-up hassles.

C. Providers

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve providers. While we certainly want providers to answer questions in this section (and in other sections) from the provider's point of view, we also invite all stakeholders to provide their viewpoints on the provider's workflows as appropriate.

1. Digital Health Apps

PR-1. What can CMS and its partners do to encourage providers, including those in rural areas, to leverage approved (see description in PC-5) digital health products for their patients?

a. What are the current obstacles?

- *Education hurdles*: Healthcare providers have to go through training programs (webinars, hands-on workshops, interactive and engaging programs). Additionally, educating patients is equally crucial
- *Cost and financial constraints*: investing in digital health tech involves upfront costs (purchasing the software, implementation, training, maintenance, hardware upgrades)
- *Data security and privacy concerns*: identity theft to the loss of patient trust
    - To prevent that, healthcare organizations need to adopt strong encryption technologies, implement access controls, regular security assessments and audits, employee training and awareness, incident response plans => **costly**
- *Resistance to change*: a complex mis of fear, uncertainty, and lack of understanding due to fear of unknown, lack of awareness (seem like an unnecessary complication rather than a valuable tool), perceived threats to job security, and importantly, comfort with current practices

b. What information should providers share with patients when using digital products in the provision of their care?

Providers should share detailed information with patients about the digital products (purpose, how they work, any potential benefits and risks); data privacy and security measures (access to the data, how it will be shared between providers or third parties, their rights regarding their data)

c. What responsibilities do providers have when recommending the use of a digital product by a patient?
- Ensure that patients clearly understand and know how to use the product as well as its specific features and functions
- Inform patients about the privacy and security safeguards in place to protect their information and their rights regarding their data (access, correct, or delete)
- Ensure that patients acknowledge any potential risks (errors, security breaches, or possibility of inaccurate information)
- Provide instructions or any necessary training or support to address any concerns or questions they may have

PR-2. What are obstacles that prevent development, deployment, or effective utilization of the most useful and innovative applications for physician workflows, such as quality measurement reporting, clinical documentation, and billing tasks? How could these obstacles be mitigated?

- Quality and safety considerations: the lack of clarity regarding AI content liability

-> in cases of errors that harm patients, currently, there is no precise line of responsibility between healthcare providers, regulators, and app developers.

- Data privacy and security: potential for data breaches and identity theft

-> healthcare organizations need robust safeguards to protect patient information from unauthorized access, breaches, or misuse

- Inequality in access: not everyone has equal access to digital devices, internet connectivity, or technology, such as low-income individuals or older adults

- System compatibility issues: apps may not be compatible with current systems, or healthcare platforms => healthcare staffs will need to be introduced and trained the new system

- Although the use of a digital product enhances convenience and efficiency for both providers and patients, many physicians question the accuracy and reliability of data collected and information

PR-3. How important is it for healthcare delivery and interoperability in urban and rural areas that all data in an EHR system be accessible for exchange, regardless of storage format (for example, scanned documents, faxed records, lab results, free text notes, structured data fields)? Please address all the following:

a. Current challenges in accessing different data formats.
    - Different EHR systems use varying formats => causing duplication of patient records, incomplete or inaccurate information
    - There is no universal standard for EHR interoperability, which can lead to confusion and impede data exchange
    - Sharing patient data across systems raises concerns about data security and privacy
    - Budget and costs: healthcare organizations and technical developers need to implement a modern interoperable health system
b. Impact on patient care quality.
    - Improved access and efficient care coordination – sharing medical records, allowing physicians to timely access to critical information for patient safety, particularly during emergencies; dealing with transportation or access issues in rural patients (elderly, low-income, or underserved populations) => receive continuous, coordinated care even on multiple healthcare providers or services
    - Patients in rural areas can access their own healthcare records and engage more actively in their care (communicate effectively with providers, schedule appointments, access lab results, and prescription orders)

- Data sharing for disease surveillance enables rural healthcare providers to report on disease trends, outbreaks, or chronic disease patterns.

c. Technical barriers to full data accessibility.
Limited resources that can prevent rural healthcare communities from full data accessibility include no fast-speed internet access and limited technologies, limited technical expertise, rural areas may use different EHR systems than larger urban hospitals => more challenging to exchange information

d. Cost or privacy implications of making all data formats interoperable.
   - Reduced healthcare costs by avoiding duplicated tests, reduced travel costs and time, low hospitalization costs, and reduced cost in manually managing and sharing data across different systems (reduced overall healthcare expenses)
   - Rural hospitals often struggle to afford the necessary technology infrastructure for EMR, ongoing maintenance fees, high standards for data privacy and security
   - In case of moving from one healthcare software platform to another, some healthcare organizations may pay at least 3-4 different types of fees: Software License fees to "turn on" certain modules, software maintenance fees for those modules, Professional service fees for the vendors to implement, and possible "per transaction" fees

e. Priority level compared to other interoperability needs.
Access to all data in EHRs, regardless of format, should be a high priority need for interoperability. A standardized, structured data exchange is foundational to system-to-system exchange to support effective workflows and safely accessing and sharing data across systems, especially during emergencies

PR-4. What changes or improvements to standards or policies might be needed for patients' third-party digital products to have access to administrative workflows, such as auto-populating intake forms, viewing provider information and schedules, and making and modifying an appointment?

ensure patient confidentiality, easily and have fast access to their health records and treatment plans, more effectively communicating with providers and nurses

2. Data Exchange

PR-5. Which of the following FHIR APIs and capabilities do you already support or utilize in your provider organization's systems, directly or through an intermediary? For each, describe the transaction model, use case, whether you use individual queries or bulk transactions, and any constraints:

a. Patient Access API.
    - Transaction Model: OAuth 2.0
    - Use Case: provide a secure mechanism for the app to obtain an access token, which is then used to access the FHIR server and retrieve the patient's data => ensures that only authorized individuals and apps can access the data
    - Query Types: authorization code and token request (making a request to the authorization server's token endpoint)
    - Constraints:
        - Secure Access Control: implementing strict access control mechanisms to limit access to sensitive data to only authorized personnel or apps
        - Strong authentication: implementing strong authentication methods like multi-factor authentication (MFA) to protect user accounts
b. Standardized API for Patient and Population Services.
    - Transaction Model: FHIR API aligned with ONC Certification Criteria

c. Provider Directory API.
d. Provider Access API.
e. Player-to-Payer API.
f. Prior Authorization API.
g. Bulk FHIR—Do you support Group ID-based access filtering for population-specific queries?
h. SMART on FHIR—Do you support both EHR-launched and standalone app access? What does the process for application deployment entail?
i. CDS Hooks (for clinical decision support integrations).

PR-6. Is TEFCA currently helping to advance provider access to health information?

a. Please provide specific examples.
    - Ensuring that a primary source of data is always available for providers for info exchange purposes, including diagnosis, prescriptions, and more
    - Reducing provider costs by reducing the need to create multiple point-to-point interfaces
    - Providing providers with a common set of security and privacy requirements to more effectively protect patients' data
b. What changes would you like to make?

- There are some limitations in accessing to medical records and requests to send records to third parties (not broad enough)
- Needs to improve on providing all the necessary data sharing for comprehensive care coordination
- More accurate and reliable patient matching is essential for effective data exchange
- Lower cost and technical barriers for small providers and technical developers

c. What other options are available outside of TEFCA?
EHealth Exchange, CommonWell Alliance, and Carequality

d. Are there redundant standards, protocols or channels or both that could be consolidated?
   - TEFCA vs. Direct Secure Messaging – direct messaging is still widely used for point-to-point messaging (referrals, for example), although TEFCA allows nationwide query-based care access

PR-7. What strategies can CMS implement to support providers in making high-quality, timely, and comprehensive healthcare data available for interoperability in the digital product ecosystem? How can the burden of increasing data availability and sharing be mitigated for providers? Are there ways that workflows or metrics that providers are already motivated to optimize for that could be reused for, or combined with, efforts needed to support interoperability?

CMS can support providers in making high-quality data available by standardizing data formats like FHIR and offering financial incentives for interoperability. To mitigate burden, they should streamline reporting and enable automated data extraction from EHRs. This can be achieved by showing providers how interoperability improves existing workflows for efficient care and accurate billing.

PR-8. What are ways CMS or partners can help with simplifying clinical quality data responsibilities of providers?

Standardizing quality measures and reporting formats across different programs and payers. They should also promote the use of automated data extraction via APIs directly from EHRs, reducing manual data collection and submission burdens on providers.

a. What would be the benefits and downsides of using Bulk FHIR data exports from EHRs to CMS to simplify clinical quality data submissions? Can CMS reduce the burden on providers by performing quality metrics calculations leveraging Bulk FHIR data exports?

Benefits include significantly reducing provider burden by automating data submission and enabling CMS to perform calculations, ensuring greater data consistency and accuracy. Downsides involve ensuring the quality and completeness of the bulk FHIR data, addressing security and privacy concerns for large datasets, and the need for robust data mapping and validation on CMS's end. Yes, CMS can reduce the burden by performing calculations if they receive standardized, high-quality Bulk FHIR data.

b. In what ways can the interoperability and quality reporting responsibilities of providers be consolidated so investments can be dually purposed?

Responsibilities can be consolidated by aligning quality measures across CMS and private payers, ensuring providers can collect data once to meet multiple reporting needs. Mandating FHIR-based APIs for both interoperability and quality reporting allows providers' IT investments to serve both data exchange and submission purposes.

c. Are there requirements CMS should consider for data registries to support digital quality measurement in a more efficient manner? Are there requirements CMS should consider for data registries that would support access to real-time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

Yes, CMS should require data registries to support FHIR-based data submission and real-time data exchange capabilities using standardized APIs. Registries should also be mandated to provide actionable, real-time feedback and performance insights directly back to providers through their EHRs, enabling immediate clinical improvements in addition to simplifying reporting.

3. Digital Identity

PR-9. How might CMS encourage providers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins that need to be tracked for each provider relationship?

Mandating their acceptance through regulations, offering financial incentives for adoption, and highlighting how a single trusted login reduces administrative burden from managing many proprietary accounts.

a. What would providers need help with to accelerate the transition to a single set of trusted digital identity credentials for the patient to keep track of, instead of one for each provider?

Clear technical guidance and implementation support for integrating these credentials into existing EHRs and workflows. They also need training for staff and assurance that adopting these systems will genuinely reduce their administrative overhead and improve efficiency.

b. How might CMS balance patient privacy with convenience and access to digital health products and services that may lead to significant improvements in health?

Implementing strong, auditable privacy-enhancing technologies alongside digital identity solutions, such as granular consent management that allows patients to control who sees what data. They must also educate patients on their data rights and the security measures in place, building trust while promoting the clear health benefits of seamless digital access.

PR-10. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs):

a. What are the challenges and benefits for providers?

For providers, digital identity credentials offer benefits like enhanced security for patient data, streamlined login processes across systems, and reduced administrative burden from managing multiple passwords. Challenges include the initial effort for integration into existing workflows and systems, and ensuring all staff are trained on their use.

b. How would requiring their use improve access to health information?

Improve secure access to health information by providing a consistent, trusted way for patients and authorized providers to log in to various health systems. This single, high-assurance identity would eliminate fragmented access issues and simplify obtaining comprehensive patient data across different healthcare organizations.

c. What are the potential downsides?

Risk of excluding individuals without easy access to technology or digital literacy, creating a single point of failure for widespread access if a system is compromised, and initial implementation costs and complexities for healthcare organizations. There are also ongoing concerns about privacy if not managed transparently.

d. What impact would mandatory credentials have on a nationwide provider directory?

Enhance the accuracy and reliability of a nationwide provider directory by ensuring provider identities are verified and consistently linked to their data. This would enable more trustworthy and real-time updates to provider information and their associated FHIR endpoints, improving its utility for all users.

e. How could digital identity implementation improve provider data flow?

Improve provider data flow by enabling secure, automated authentication for system-to-system communication and API-based data exchange, removing manual verification steps. It fosters trust between data-sharing entities, allowing for more seamless and efficient transfer of patient information between disparate systems.

f. Would combining FHIR addresses and identity improve data flow?

Yes, it ensures that when a system requests data from a FHIR endpoint, the identity of the requester is securely and consistently verified, allowing for trusted, targeted, and compliant exchange of specific health information.

PR-11. How could members of trust communities [3] (for example, QHINs, participants and sub participants in TEFCA, which requires Identity Assurance Level 2 (IAL2) via Credential Service Providers (CSPs)) better support the goals of reduced provider and patient burden while also enhancing identity management and security?

Trust communities can reduce provider and patient burden while enhancing identity management and security by fully leveraging standardized IAL2 digital identity credentials for seamless, multi-system access. This means consistently accepting one trusted login for patients and providers across all participating entities, eliminating proprietary logins and redundant identity verification steps. By doing so, they not only streamline workflows and reduce password fatigue but also significantly bolster security by relying on high-assurance identity proofing for all data exchanges.

4. Information Blocking

PR-12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?

Yes, ASTP/ONC should consider revising or removing some information blocking exceptions. The main goal would be to make it much easier for patients and other providers to access and share electronic health information, which also encourages more competition among healthcare services.

PR-13. For any category of healthcare provider (as defined in 42 U.S.C. 300jj(3)), without a current information blocking disincentive established by CMS, what would be the most effective disincentive for that category of provider?

For healthcare providers not yet facing penalties for information blocking, the most effective disincentive would likely be tying compliance directly to their revenue or program participation. This means either implementing financial penalties or reducing their eligibility for federal programs like Medicare or Medicaid. Hitting their bottom line or ability to operate within key programs would strongly incentivize them to share patient data.

PR-14. How can CMS encourage providers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Make the reporting process clear, easy, and secure. They should also highlight how reporting helps create a fairer data-sharing environment for everyone. This would significantly advance data exchange by holding those who block information accountable, ultimately improving patient care and promoting healthier competition within the healthcare system.

D. Payers

PA-1. What policy or technical limitations do you see in TEFCA? What changes would you suggest to address those limitations? To what degree do you expect these limitations to hinder participation in TEFCA?

Voluntary participation and managing varied privacy rules across states are possible limitations. These limitations could slow down widespread adoption, as providers might wait for stronger incentives or clearer technical guidance before fully committing.

PA-2. How can CMS encourage payers to accelerate the implementation and utilization of APIs for patients, providers, and other payers, similar to the Blue Button 2.0 and Data at the Point of Care APIs released by CMS?

Continue to mandate use through regulations, with clear deadlines and enforcement. Offer incentives for early adopters or those exceeding basic requirements, alongside providing robust technical guidance and support to ease the implementation burden.

PA-3. How can CMS encourage payers to accept digital identity credentials (for example, CLEAR, ID.me, Login.gov) from patients and their partners instead of proprietary logins?

Make it a rule for them to accept these trusted logins instead of their own. Offer money or rewards to payers who adopt them quickly, and provide clear instructions and support to make implementation easier. CMS should lead by example by fully using these digital IDs themselves for Medicare services.

PA-4. What would be the value to payers of a nationwide provider directory that included FHIR end points and used digital identity credentials?

It would drastically reduce administrative costs and errors by ensuring highly accurate, real-time provider information for claims and prior authorizations. This improved data quality and accessibility would also streamline operations, leading to more efficient processes and better experiences for their members who can easily find accurate in-network providers.

PA-5. What are ways payers can help with simplifying clinical quality data responsibilities of providers?

Standardizing data collection and reporting requirements across different health plans, reducing varied requests. Leverage technology like FHIR-based APIs to automate data exchange directly from providers' electronic health records.Provide actionable insights and pre-populated data to providers, reducing the manual burden of data aggregation and helping them focus on care improvement.

a. How interested are payers and providers in EHR technology advances that enable bulk extraction of clinical quality data from EHRs to payers to allow them to do the calculations instead of the provider-side technology?

Providers are highly interested in offloading the complex and burdensome process of calculating quality measures, which currently consumes significant time and resources. Payers are interested in receiving this data in bulk to improve accuracy, gain deeper insights for value-based care, and reduce their own administrative costs associated with manual data collection and validation.

b. In what ways can the interoperability and quality reporting responsibilities of providers to both CMS and other payers be consolidated so investments can be dually purposed? Are there technologies payers might leverage that would support access to real time quality data for healthcare providers to inform clinical care in addition to simplifying reporting processes?

CMS and other payers can work together to standardize quality measures and reporting formats, ensuring providers collect data once for multiple purposes. Technologies like FHIR-based APIs and bulk data access can then be used to automate the extraction of this data directly from EHRs, allowing payers to perform calculations. This not only simplifies providers' reporting burden but also provides payers with real-time quality data that can then be shared back with providers to inform clinical decision-making.

PA-7. How can CMS encourage payers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would it advance or negatively impact data exchange?

Encourage payers to submit information blocking complaints by providing clear, easy-to-use channels and publicly acknowledging the importance of their input in ensuring data flow.

E. Technology Vendors, Data Providers, and Networks

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve technology vendors, data providers, and networks. While we certainly want technology vendors, data providers, and networks to answer questions in this section (and in other sections) from their point of view, we also invite all stakeholders to provide their viewpoints on the technology vendor, data provider, and network use cases as appropriate.

1. Ecosystem

TD-1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?

Simplify reimbursement for digital health and expand access to Medicare data via FHIR APIs. For longer-term growth, establish stable, adequate payment policies for digital tools, gather strong evidence of their value, and invest in foundational digital infrastructure like modern identity verification.

TD-2. Regarding CMS Data, to stimulate developer interest—

a. What additional data would be most valuable if made available through CMS APIs?

Comprehensive, de-identified clinical data (beyond just claims), detailed provider quality and performance metrics, and social determinants of health (SDOH) data, medication management.

b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?

Clinical data from Electronic Health Records (EHRs), including diagnoses, lab results, medications, and clinical notes, Patient-Generated Health Data (PGHD) from wearables, home monitoring devices, and patient apps would provide valuable data.

c. What obstacles prevent accessing these data sources today?

Interoperability challenges between disparate EHR systems, as they often use different formats and lack seamless data exchange capabilities, privacy concerns and complex regulatory requirements, such as HIPAA, also restrict data flow, making secure and compliant sharing difficult. Lack of standardization for PGHD and SDOH data, along with workflow integration issues for providers to collect and manage this diverse information, significantly prevent comprehensive access and utilization.

d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?

CMS and ASTP/ONC should include APIs for Patient-Generated Health Data, Social Determinants of Health, advanced clinical details, and public health reporting in their program policies to truly unleash innovation and better support patients and providers.

2. Digital Identity

TD-3. Regarding digital identity implementation:

a. What are the challenges and benefits?

Implementing digital identity credentials in healthcare offers key benefits like enhanced security and streamlined access to health information, making it easier for patients and caregivers to manage their data securely. However, significant challenges include the complex signup process which can exclude some users, privacy concerns about data sharing, and the ongoing need for widespread adoption across all healthcare providers.

b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800-63-3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?

It would provide stronger, more reliable identity verification for accessing sensitive health data. For data exchange, it would streamline and secure interactions between different systems.

c. What impact would mandatory use of the OpenID Connect identity protocol have?

This would standardize and secure user authentication across various systems and enable seamless single sign-on (SSO) for patients and providers accessing different health applications, reducing login fatigue and improving user experience. It would also enhance interoperability by providing a consistent and trusted way to verify identities, which is essential for secure health data exchange.

3. Technical Standards and Certification

TD-4. How can CMS better encourage use of open, standards-based, publicly available APIs over proprietary APIs?

Strengthening mandates and enforcement for adoption across various payer and provider types. Also provide clear, consistent technical guidance and implementation resources to reduce development burden.

TD-5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?
A nationwide provider directory with FHIR endpoints would vastly improve health information access for patients, providers, and payers by offering accurate, real-time data for better care coordination and reduced administrative burden. A neutral, government-backed entity like ONC or CMS should publish it, and users should not bear a direct cost to ensure widespread public benefit.

TD-6. What unique interoperability functions does TEFCA perform?

It establishes a universal policy and technical framework for health information exchange nationwide, binding diverse networks (called QHINs) and their participants to common rules. This enables seamless data sharing across different organizations that were previously disconnected, especially for specific purposes like treatment, payment, public health, and individual access

a. What existing alternatives should be considered?

Fragmented patient portals, direct EHR vendor APIs, and established regional Health Information Exchanges (HIEs)

b. Are there redundant standards, protocols or channels or both that should be consolidated?

TD-7. To what degree has USCDI improved interoperability and exchange and what are its limitations?

It has improved interoperability by standardizing a minimum set of health data elements for consistent exchange, boosting patient access and reporting. However, its limitation as a minimum dataset means much comprehensive health data remains outside its scope, and adoption of new versions can lag.

a. Does it contain the full extent of data elements you need?

Sure

b. If not, is it because of limitations in the definition of the USCDI format or the way it is utilized?
c. If so, would adding more data elements to USCDI add value or create scoping challenges? How could such challenges be addressed?
d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?

No

TD-8. What are the most effective certification criteria and standards under the ONC Health IT Certification Program?

The most effective ONC certification criteria are those mandating FHIR-based APIs and the United States Core Data for Interoperability (USCDI), which improve interoperability and patient access to health data. Criteria promoting algorithmic transparency for clinical decision support are increasingly vital for responsible health IT use.

TD-9. Regarding certification of health IT:

a. What are the benefits of redefining certification to prioritize API-enabled capabilities over software functionality?

Prioritizing APIs would enhance interoperability, allowing different systems and apps to exchange data more seamlessly and giving patients more choice. It also helps future-proof health IT by ensuring data can flow flexibly as new technologies emerge.

b. What would be the drawbacks?

Increased complexity for developers needing to conform to strict API standards, potential for security vulnerabilities if APIs are not rigorously secured, and the risk of misinterpreting or misusing data if not properly contextualized through the API. It could also initially raise implementation costs for some vendors.

c. How could ASTP/ONC revise health IT certification criteria to require APIs to consistently support exchanging data from all aspects of the patient's chart (for example, faxed records, free text, discrete data)?

ASTP/ONC could revise criteria by mandating standardized data models for unstructured data (like faxed or free text notes, perhaps using clinical natural language processing standards) and requiring specific FHIR resources or extensions that can reliably carry these data types alongside discrete data. They would need to define precise technical

requirements for how *all* types of patient information, once digitized, are made available via APIs.

d. What policy changes could CMS make so providers are motivated to respond to API-based data requests with best possible coverage and quality of data?

Implement performance-based incentives or disincentives, tying provider reimbursement or participation in certain programs to the quality and completeness of data shared via APIs. Require attestation or audits of API data quality, while simultaneously providing resources and technical assistance to help providers meet these standards.

e. How could EHRs capable of bulk data transfer be used to reduce the burden on providers for reporting quality performance data to CMS? What capabilities are needed to show benefit? What concerns are there with this approach?

EHR bulk transfer can automate quality reporting, significantly reducing provider burden. This requires standardized export and secure transmission. Concerns include data quality, security, and provider control over the transferred information.

TD-10. For EHR and other developers subject to the ONC Health IT Certification Program, what further steps should ASTP/ONC consider to implement the 21st Century Cures Act's API condition of certification (42 U.S.C. 300jj-11(c)(5)(D)(iv)) that requires a developer's APIs to allow health information to be accessed, exchanged, and used without special effort, including providing access to all data elements of a patient's electronic health record to the extent permissible under applicable privacy laws?

Increase auditing and enforcement to ensure certified APIs are actually delivering on this promise in real-world scenarios, including assessing the ease of app registration and user onboarding. Furthermore, encouraging industry best practices for standardized API documentation and testing tools would help developers build and implement compliant systems more effectively.

TD-11. As of January 1, 2024, many health IT developers with products certified through the ONC Health IT Certification Program are required to include the capability to perform an electronic health information export or "EHI export" for a single patient as well as for patient populations (45 CFR 170.315(b)(10)). Such health IT developers are also required to publicly describe the format of the EHI export. Notably, how EHI export was accomplished was left entirely to the health IT developer. Now that this capability has been in production for over a year, CMS and ASTP/ONC seek input on the following:

a. Should this capability be revised to specify standardized API requirements for EHI export?

Yes, the EHI export capability should be revised to specify standardized API requirements. This would ensure consistency, improve interoperability, and reduce fragmentation in how data is exported. A standardized approach would simplify data exchange for patients, providers, and other developers, ultimately making the EHI export more usable and valuable.

b. Are there specific workflow aspects that could be improved?

Workflow improvements could focus on making the EHI export process more intuitive and less burdensome for providers and patients. This includes streamlining the initiation of an export, providing clearer status updates, and ensuring that the exported data is easily understandable and usable by the recipient. Automating the notification and delivery of the export, while maintaining security, would also be beneficial.

c. Should CMS consider policy changes to support this capability's use?

Yes, CMS should consider policy changes to actively support and incentivize the use of standardized EHI export capabilities. This could involve tying participation in certain programs or quality measures to the successful and routine use of EHI export, or offering incentives for providers who demonstrate high rates of EHI export utilization. Policies could also focus on educating patients and providers about the availability and benefits of this capability

4. Data Exchange

TD-12. Should CMS endorse non-CMS data sources and networks, and if so, what criteria or metrics should CMS consider? If CMS were to endorse non-CMS data sources, this would improve accessibility and help with continuity of care. Data privacy and security could become an issue that needs to be addressed.

TD-13. What new opportunities and advancements could emerge with APIs providing access to the entirety of a patient's electronic health information (EHI)?

a. What are the primary obstacles to this? The primary obstacles include: security and privacy concerns, data availability and access, and lack of standards and business workflow.
b. What are the primary tradeoffs between USCDI and full EHI, especially given more flexible data processing capabilities today? USCDI is easier to use and patient/provider helpful. Full EHI is more comprehensive and provides more information.

TD-14. Regarding networks' use of FHIR APIs:

a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?
b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?
c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?

TD-15. Regarding bulk FHIR APIs:

a. How would increased use of bulk FHIR improve use cases and data flow?

Enabling the efficient, standardized transfer of large datasets for entire patient populations in a single request, rather than individual records. This capability vastly enhances use cases like population health management, quality reporting, research, and value-based care analytics, making data collection and aggregation much faster and less burdensome.

b. What are the potential disadvantages of their use?

Potential disadvantages include ensuring the quality and completeness of these large datasets, managing the security and privacy risks associated with transferring vast amounts of sensitive information, and addressing the technical complexities of implementation and robust error handling for such large-scale exchanges. Initial performance limitations in some EHR implementations also remains a concern.

TD-16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?

Maintaining point-to-point data exchange models offers simplicity for a few connections but quickly becomes complex, costly, and difficult to manage as the number of connections grows. In contrast, a shared network infrastructure (like TEFCA's network of networks) provides scalability, standardized data exchange, and centralized governance, significantly reducing individual integration burdens and fostering broader interoperability

a. Do current rules encourage scalable network participation?

Currently, rules like the 21st Century Cures Act and TEFCA aim to encourage scalable network participation by mandating open APIs and creating a common framework for

trusted exchange. However, the voluntary nature of full participation in some aspects can still limit reach.

b. What changes would improve alignment (for example, API unification, reciprocal access)?

To improve alignment, unifying API standards (e.g., universal FHIR adoption) and promoting true reciprocal access where data flows freely and consistently in both directions are crucial. This means moving beyond one-off data pulls to a system where all connected parties can easily access and exchange information based on common rules and trust.

TD-17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?

They should establish policies and funding mechanisms that incentivize participation and sustain operations for Qualified Health Information Networks (QHINs) and other exchange entities. This includes defining clear value propositions for sharing, potentially linking participation to quality programs or reimbursement models. Their role also extends to setting and enforcing standards (like FHIR and USCDI) that reduce technical barriers and encourage a balanced supply and demand for data exchange, ultimately driving its widespread use and positive health outcomes.

5. Compliance

TD-18. Information blocking:

a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non-responsiveness as well as situations that may cause a failure or unusable response? Examples of information blocking include: charging excessive fees, changing the length of time a patient has access to records, or refusing to transfer PHI to a third party.
b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices? The following could discourage healthcare providers from engaging in information blocking practices: enhanced surveillance, disclosure of restrictions, transparency on fees, and clarifying information blocking exceptions.
c. Are there specific categories of healthcare actors covered under the definition of information blocking in section 3022(a)(1) of the Public Health Service Act (PHSA) that lack information blocking disincentives? These are the healthcare actors

covered under the definition of information blocking: healthcare providers, health IT developers, and HINs/HIEs.

TD-19. Regarding price transparency implementation:

a. What are current shortcomings in content, format, delivery, and timeliness? Timeliness, inaccuracy and incorrect information, and website and data issues are shortcomings for price transparency.
b. Which workflows would benefit most from functional price transparency? The workflows that would benefit most from functional price transparency include: patient care and decision making, healthcare providers workflow, and business operations.
c. What improvements would be most valuable for patients, providers, or payers, including CMS? Being transparent with prices helps patients and providers make the best decisions regarding healthcare. This will help decrease stress and conflict amongst these individuals.
d. What would further motivate solution development? If providers and patients start to see improvements in the process and system, this will be more encouraging. If value-based care starts to show effectiveness in improving patient outcomes, this will motivate solution development.

F. Value-Based Care Organizations

This section is intended for all stakeholders to provide input on questions as they relate to use cases and workflows that involve value-based care organizations. While we certainly want value-based care organizations to answer questions in this section (and in other sections) from the value-based care provider point of view, we also invite all stakeholders to provide their viewpoints on the value-based care workflows as appropriate.

1. Digital Health Adoption

VB-1. What incentives could encourage APMs such as accountable care organizations (ACOs) or participants in Medicare Shared Savings Program (MSSP) to leverage digital health management and care navigation products more often and more effectively with their patients? What are the current obstacles preventing broader digital product adoption for patients in ACOs?

When ACOs and MSSP programs utilize digital health management and care navigation products with more success, then they can be reimbursed or given money since this is a value-based care model. These patients who are receiving care through Medicare and Medicaid do not have access to certain forms of digital products. Not all of their healthcare information is under one system, which would be difficult to collect into one.

VB-2. How can key themes and technologies such as artificial intelligence, population health analytics, risk stratification, care coordination, usability, quality measurement, and patient engagement be better integrated into APM requirements?

AI can be used to help create payment plans for patients who have specific criteria. AI can be used to gather data to identify high risk patients and how to address specific social determinants of health. Promoting team-based care by requiring interprofessional communication could help patients receive quality care.

VB-3. What are essential health IT capabilities for value-based care arrangements?

a. Examples (not comprehensive) may include: care planning, patient event notification, data extraction/normalization, quality performance measurement, access to claims data, attribution and patient ID matching, remote device interoperability, or other patient empowerment tools.
b. What other health IT capabilities have proven valuable to succeeding in value-based care arrangements?

Essential health IT capabilities for value-based care arrangements include: data analytics and reporting, care coordination tools, patient portals and apps, streamlined administrative processes, support for alternative payment methods, and population health management.

VB-4. What are the essential data types needed for successful participation in value-based care arrangements?

The four essential types of data needed for successful participation in value-based care include patient data, clinical quality data, cost and utilization data, and administrative and contractual data.

2. Compliance and Certification

VB-5. In your experience, how do current certification criteria and standards incorporated into the ONC Health IT Certification Program support value-based care delivery?

The ONC Health IT Certification Program allows every system that utilizes this program the ability to share information. The program can accurately capture quality measures to help ensure healthcare systems are receiving their payments. Patient portals are utilized to improve patient engagement.

VB-6. What specific health information technology capabilities that could benefit APMs are not currently addressed by existing certification criteria and standards that should be included under the ONC Health IT Certification Program?

Areas not addressed by existing certification criteria include: advanced data analytics, support for complex management workflows, robust integration with social determinants of health information, and standardized reporting for APM performance.

VB-7. How can technology requirements for APMs, established through CEHRT or other pathways, reduce complexity while preserving necessary flexibility?

To reduce complexity, the following could be done: streamlined data collection and reporting, decreased duplicate reporting, and easier compliance. To preserve flexibility, the following could be done: focus on interoperability, allowing for new technologies and versions, and support for a diverse technology solutions.

VB-8. How can other HHS policies supplement CEHRT requirements to better optimize the use of digital health products in APMs? As an example, requirements under the Conditions of Participation for hospitals (42 CFR 482.24(d)) require hospitals to transmit electronic patient event notifications to community providers. What barriers are in place preventing APM participants from receiving the same notifications?

CEHRT should combine with other HHS policies that focus on interoperability and reduction of information blocking to optimize the use of digital health products. Not having these has made it difficult for APM participants to receive the same notifications.

VB-9. What technology requirements should be different for APM organizations when comparing to non-APM organizations (for example, quality reporting, and interoperability)?

The following should be required for technology: detailed data collection and reporting, sophisticated data analytics, seamless data exchange and interoperability, and advanced EHR functionality.

VB-10. In the Calendar Year (CY) 2024 Physician Fee Schedule final rule (88 FR 79413), CMS established that CEHRT requirements for Advanced APMs beyond those in the "Base EHR" definition should be flexible based on what is applicable to the APM that year based on the area of clinical practice. What certification criteria should CMS identify under this flexibility for specific Advanced APMs, or for Advanced APMs in general? Are there specific flexibilities or alternatives to consider for smaller or resource-constrained (such as rural) providers in meeting CEHRT requirements without compromising quality of care or availability of performance data?

These are the specific criteria CMS should focus on for flexibility: EHRs focused on clinical area-specific, supportive of interoperability, relevant to quality measurement, and aligned with APM objectives. The following are ways to provide support: financial, gradual integration, group reporting options, and enhanced technical assistance.

3. Technical Standards

VB-11. What specific interoperability challenges have you encountered in implementing value-based care programs?

The following are challenges faced in implementing value-based care programs: patient data spread across several health systems, financial risks, providers are reluctant to switch, and social determinants of health.

VB-12. What technology standardization would preserve program-specific flexibility while promoting innovation in APM technology implementation?

By implementing open standardization for technology, this will allow providers and patients to provide information to one another and allow continuation of care. All healthcare providers should be able to access the same information to provide the best care for patients.

VB-13. What improvements to existing criteria and standards would better support value-based care capabilities while reducing provider burden?

Ways to improve the provider burden is to make prior authorizations a streamlined process will help this immensely.

VB-14. How could implementing digital identity credentials improve value-based care delivery and outcomes?

Digital identity credentials provide patients with easier access to their health information. This will decrease the amount of requests for information and allow providers to see information that is useful in a quick manner.

VB-15. How could a nationwide provider directory of FHIR endpoints help improve access to patient data and understanding of claims data sources? What key data elements would be necessary in a nationwide FHIR endpoints directory to maximize its effectiveness?

A nationwide provider directory of FHIR endpoints would consolidate data, which would include claims and patient information. This would streamline the process of providing care to patients significantly. More information that should be included in a FHIR system includes: provider identifiers, guidelines on how information is used, labs, and MARs.